Block 7

Financial Stability

UNIT 24 MACROECONOMICS, FINANCE AND BUSINESS CYCLES

Structure

24.0 OBJECTIVES

After reading this Unit, you should be able to:

- identify macroeconomic issues in an economy;
- identify the examples of business cycles;
- explain the functioning of the financial system;
- identify asymmetric information and its effects;
- define financial crisis; and
- explain the linkages between the financial crisis and working of the economy.

24.1 INTRODUCTION

As discussed in previous blocks, the financial system plays an important role in the economy and has always been an interesting subject for economists and policymakers. In the present-day world, the financial sector of a country is very important not only for advancement or growth but also for the stability of the economy. We have discussed in Unit 1 that the financial system includes three key elements: (i) financial institutions, (ii) financial markets, and (iii) financial instruments. The major function of the financial system in an economy is to work as an intermediary between lenders (or, savers) and borrowers (or, investors). Governments, corporations/business firms and individuals borrow from the financial markets to fulfil their objectives and invest their savings in financial instruments to grow their wealth. These decisions, in turn, affect macroeconomic variables like aggregate demand, output, employment, and prices. Macroeconomic variables affect financial conditions in a country and the developments in the financial sector are important for capital formation which is a prerequisite for the growth and development of the economy. However, we have also seen that at times, the developments in the financial sector leads to macroeconomic instability. The memory of the financial crisis of 2007-08 in the USA which spread across the globe within a short span of time, which is also known as the Great Financial Crisis or the Great Crash, and its after-effects on the world economy are very fresh. In this unit, we will extend our understanding of financial systems and examine the interconnection between Macroeconomics, finance and business cycles.

24.2 MACROECONOMICS AND BUSINESS CYCLES

In simple terms, macroeconomics is the study of the economy as a whole, i.e., how the economy works. It is focused on important issues like employment, inflation, output or GDP, growth rate, exchange rate, balance of payments, and business cycles. In short, we can say that it deals with the most important economic issues and problems of society. These important issues are dealt with mainly by two questions. One, how is the GDP of a country determined in the short run and the long run? Second, how is GDP (or the process of GDP determination) affected by macroeconomic variables such as consumption, investment, inflation and unemployment?

One of the major factors responsible for the growth of an economy is capital formation. Over the years, many financial innovations have taken place which have helped not only in building a capital base of the economy but also in the expansion of markets and wealth. But, the process or path of economic growth is not smooth or stable. Economies of different countries have experienced wide fluctuations in variables like employment, output, and prices over the past three centuries. Various kinds of explanations have been provided for economic fluctuations or business cycles ranging from the lack of aggregate demand to real business cycle models. Depending upon the kind of explanation, solutions for stabilising the economy have also been suggested. It has also been experienced that business cycles are often related to problems in the financial sector of the economy.

Business Cycles: It refers to ups and downs in economic activities namely, output, unemployment, and income among other things. The business cycle is an indication of how an economy is doing. A business cycle has four different stages predominantly. The first stage of a business cycle is an economic expansion. It is when the investment is increasing due to lower interest rates, output increases, and employment increases. The second stage is when the economic activities are at their peak and thus, the economy is doing extremely well. Often, it is considered that if there is a boom there is going to be a burst eventually. So, in short, in the first stage of a business cycle, the output is high, demand is high, income is high, unemployment is less, consumption is high, and investment is high. The second stage is when the economic activities are at their peak and thus, the economy is doing extremely well. In the third stage of a business cycle, aggregate demand goes down, the output starts to slow down, investment slows down, employment falls, income falls, and thus consumption also decreases. In the fourth stage of the business cycle, the condition is worse and a depression follows. Aggregate demand further decreases, investment decreases, production decreases again, and more layoffs happen.

The business cycle acts as a vicious cycle. Starting from a boom to a recession and then to a depression. One thing leads to another. For example, a lack of demand in one sector might lead to a reduction in investment and production which leads to loss of employment, a reduction in income and again a decrease in aggregate demand.

24.2.1 Case Study: The Great Depression (1929-1941)

In the 1920s the growth rate in the US was high (4.7%) and the unemployment rate was very low (3.2%). There was an exorbitant growth in wealth. With this high wealth, the investment by the public increased in the stock market. The regulations were weak in the stock market which had led to higher speculation in the market. Eventually, the stock market crashed in 1929. On October 24, 1929, the market opened down by 11% from the previous day. This is known as the 'Black Thursday'. In the following days, the market fell further. From 1929 to 1932, the stock market decreased by 85%. During the peak of the great depression, the unemployment rate was as high as 24%. A lot of blame was put on banks that had advanced too many loans by decreasing their reserves gradually. It was coupled with the expansionary monetary policy (increased money supply) which led to increased levels of borrowing. After the stock market crash instead of decreasing the rate of interest the Federal Reserve did the opposite and the rate of interest was increased. The approach was that the troubled banks would fall and a strong system would emerge. The increase in interest rate led to deflation (the opposite of inflation, prices decrease). At the same time higher tariff was imposed on agricultural imports to protect agriculture in the US which further contributed to the great depression. In 1933, President Roosevelt increased government spending but it was too late by then. Within the four years of the recession, the GDP of the US decreased by 29%.

Unemployment was at the highest (24%) level in 1933 when 1 out of every 4 people was unemployed. 30% of the banks failed. The standard of living decreased. Many factories had shut down. Rural poverty increased rapidly. The birth rate declined, and women's participation in the workforce increased. The great depression eventually ended in 1941.

24.3 FINANCE AND ECONOMY

We are generally faced with two kinds of situations. First, all of our income is not consumed and some of it is saved. For these savings, we look for options to earn some return, i.e., to invest our savings productively. Second, some people have business ideas or they have to do some transaction but do not have sufficient funds, i.e., they need funds. In macroeconomics, savings and investment are very important concepts and help in understanding macroeconomic problems like growth, inflation and unemployment and their solutions. Investment in physical and human capital is very important for a country's long-run growth. At the same time, it affects the macroeconomic stability, inflation and unemployment, in the economy. Savings are the most important component of funding investment, but every increase in savings may not be good for the health of the economy. It can lead to a shortage of aggregate demand in the economy. Moreover, it is also important how these savings are allocated to different investment options. Some options can be simply better than others in terms of productivity, outcome and impact on employment, inflation and other economic variables. The financial system of a country is generally helpful in providing solutions for the issues mentioned above. It provides productive investment options where savings can be invested and at the same time, the people in need get the opportunity to borrow the desired funds. It provides an efficient way to allocate the funds among different uses.

The financial sector is considered to be different from the real economy. The real economy produces goods and services with the help of the factors of production, land, labour, and capital. It provides us with consumption goods like food, furniture, clothes, and capital goods like plant, machinery, transportation, etc. What happens in the real economy determines an economy's capacity to produce and consume in the long run, and also affects several variables like inflation, unemployment, and distribution of income and wealth which in turn affect the economy itself. On the other hand, the financial sector provides the mechanisms and channels through which the lenders and borrowers coordinate in the market and facilitate transactions in the real economy. Efficient working of the financial system is very much desirable to increase the productivity in the economy, so that, the funds can be directed to those activities which are most productive. It is usually thought to be limited to the movement of funds in the economy to help the real economy. We can see that employment in this sector is increasing day by day. The financial system of a country not only channels the funds in the right direction but in doing so generates well-paid jobs and builds wealth for many which in turn would have an impact on activities like consumption and production. It is often a matter of debate whether the financial sector helps in the development of the economy or if it is a result of economic development, both of these arguments are true.

24.3.1 Financial System

The financial sector, today, is a very important sector for almost all economies. The major function of this sector is to provide channels for the movement of funds between borrowers and lenders. Lenders and borrowers can be individuals, households, business firms, governments and foreigners. Lenders and borrowers can interact through organised or formal financial markets or outside these markets. The transactions done outside the organised financial markets are, generally, riskier and costlier as compared to the formal markets. The financial system of a country helps in reducing the transaction costs and risks involved in financial transactions. As a country develops, its financial market also develops along with it and the proportion of the transactions outside formal financial markets declines with economic development.

The financial system includes financial institutions like banks, insurance companies, mutual funds etc. and financial markets like stock markets, bond markets etc. In financial markets, lenders can directly provide funds directly to the borrowers. For example, suppose Reliance Industries wants to expand its business and for that purpose needs some funds. It can directly sell bonds to individuals in the bond market or it can bring an IPO (initial public offering) in the stock market which can be subscribed by individuals, banks, and companies. This is **direct finance**. In this case, the individuals or the companies who have savings are directly lending their money to the borrower or buying stocks of a company. This requires a lot of information and expertise to deal directly with the financial markets. On the other hand, instead of lending directly to the borrower, an individual can do it through some intermediary. For example, I have some savings and I want to invest these savings profitably. I can deposit my money in a bank, I can invest in a mutual fund, I can buy some insurance, or I can participate in some pension scheme. Financial institutions like banks, mutual funds, insurance funds or pension funds receive money from the lenders and lend it to the borrowers. This is **indirect finance**.

In financial markets, there are different kinds of risks present for individuals, firms, and financial intermediaries. There is a risk that the deposits or the credit given may not be paid back partially or fully, known as ***credit or default risk***. Financial intermediaries including banks may find themselves unable to meet the liquidity demands due to sudden and unexpected increases in the demand for liability payments or withdrawal of deposits, known as ***liquidity risk***. Volatility in the interest rates and asset prices can deteriorate the financial health or the balance sheets, known as ***interest rate and price***

risk. Volatility in the exchange rate affects the value of assets and liabilities held in foreign currency and may worsen the balance sheets, known as the ***exchange rate risk***. Due to one or more risks mentioned above, the value of assets of financial intermediaries, or firms and individuals, may fall below liabilities leading to ***insolvency***. One of the important reasons for the risk is **asymmetric information**. One of the important aspects of the functions of financial intermediaries is to assess and manage these risks.

24.3.2 Asymmetric Information, Adverse Selection and Moral Hazard

When, in a transaction, one party does not have the same information as the other party it is known as ***asymmetric information***. For example, suppose there is a firm which is planning to undertake a new project and for this project firm borrows from the market. The firm has different assessments for the risks involved in the project, and prospects of the future return. But, while approaching the lenders or investors this firm may not reveal all the information related to the project it has. In this case, the lenders/investors would have less information than the borrower.

It happens at times that we try to hide some information which may impede our chances to borrow from the market. Suppose, an individual or a firm is willing to undertake a very risky project and lenders or investors may not be willing to take that kind of risk. To get the necessary funding the borrower may hide facts or information about the risk and make best efforts to get the loan, and may even offer to pay a higher interest rate. If the lenders/investors, pursued by the borrower, invest in the project without knowing the risk it creates ***an adverse selection problem***. Due to the risky nature of the project, the borrower may not be able to honour its liabilities, i.e., the risk of default. There is another problem that arises because of asymmetric information which is known as ***moral hazard***. This may occur irrespective of whether there is an adverse selection or not; this takes place after the financial transaction has occurred. A borrower may divert the borrowed funds to some risky project or activity. For example, the firm may put its investment in risky bonds that might pay or not pay. If these risky bonds pay then the bank will get its money back and if not, then the firm will not be able to pay back and most likely will default. Another simple example of moral hazard is that suppose you have taken insurance for losses due to fire. After taking the insurance you may decide not to put in place an efficiently working fire safety system to cut down on costs.

24.3.3 Case Study: Satyam Computers Scandal

Satyam Computer Services Limited (popularly known as Satyam Computers), a company established in 1987, was the fourth biggest Indian company in the IT sector in the 2000s. It was one of the finest examples of India's growth story. From 2003 to 2008, the annual growth rate of the company was 35 per cent. In January 2009, the Chairman and the founder of the company confessed that he had been manipulating the accounts of the company. He inflated the balance sheet and showed the profits and cash which never existed. This false growth also inflated the stock prices of the company. This was done to obtain cheaper loans which were diverted to the real estate. The company's auditors turned a blind eye to the wrongdoings. The company lost its customers, stock prices plumped, and investors lost billions. The company was acquired by Tech Mahindra in April 2009. In Indian corporate history, the Satyam scandal is a very fine example of corporate fraud. This highlights the problem of asymmetric information that led to adverse selection due to false information and moral hazard on the part of managers and auditors. This episode also highlights the importance of prudent regulation that would help in decreasing the information asymmetry and promote transparency.

Check Your Progress 1

1) Discuss different stages of business cycles.

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2) How important is the concept of asymmetric information for the financial sector?

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3) What are the different kinds of risk associated with financial transactions?

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24.4 FINANCIAL CRISIS

The financial crisis can be understood as an upset in the financial markets in the form of wide fluctuations in asset prices like stocks or real estate. It is associated with fluctuations in the flow of credit and disruptions in the working of financial intermediaries and markets. Financial stability helps in better assessment and management of financial risk and thus helps in the efficient allocation of resources which in turn, promotes economic growth. Although, with financial innovations, the understanding of risk and its management has also improved it is not possible to avoid risk completely because of the uncertainty.

The financial system is like a complex web of interactions between different players. For example, person A has invested money in Mutual Fund X, this mutual fund has invested money in another investment fund which has further invested money in a bank that has given a loan to a corporate firm. You have a financial claim over a part of the mutual fund. The amount of money you can get by selling your holding of the mutual fund depends upon many things. First, the price or value of the fund may be affected by the conditions of the market. Second, the value of underlying assets may further be a function of different variables. And third, if the mutual fund in question has sufficient liquidity. In financial markets, whether the contract between two parties will be honoured or not depends upon other parties who are linked to this contract by a chain of financial transactions. A disturbance at any point in this chain of transactions can cause financial turbulence in the market and it can be a reason for a **financial crisis.** Speculative activities in the financial markets along with asymmetric information are often the sources of financial fluctuations. There may be several other factors that may start or may increase the intensity of a financial crisis.

It is not so that all the financial crises are the same in the nature and magnitude but they have some common elements. Usually, a financial crisis starts with an asset price boom. These elements are described in the following paragraphs:

Asset Price Booms and Busts: At times, the prices of financial assets like stocks or real estate properties increase by a large amount over their **fundamental values**. This is known as **the asset-price bubble.** Often, this is led by the optimism of the investors about the future value of the asset. This may also happen because of a significant increase in the availability of credit. With the easy availability of credit, investors may increase their investment in an asset they expect to give very good returns in the future. This kind of increase can be termed a **credit boom** or **credit bubble**. The optimism about the asset prices and credit boom can have a feedback effect on each other, which may further intensify the asset-price bubble. The asset price booms cannot last forever, even a small shock in the market can lead to the bust of the bubble. For example, if investors start to feel the future rate of return for the asset may be lower, they may start to withdraw their investment which will lead to a decline in the asset prices.

Insolvencies of the financial institutions: At times, the loans given by the banks are secured by the collateral or mortgage of the asset for which the loan has been provided. When the prices of such assets decline, it will create problems for the lending banks. In case of defaults by the borrowers, the banks will not be able to recover their loans. This problem is further aggravated by the **leveraging** by the lending institutions. In an optimistic environment, to increase the amount of loans/credit banks often borrow from other financial institutions. The borrowed funds increase banks' capacity to lend. This process is known as leveraging. But in case of asset-price bust, the value of mortgage declines and banks are unable to recover their loans and thus, fall short of meeting their payment obligations. In this situation, the capital of the banks falls and they reduce their lending. Which may lead to a further decrease in asset prices. The value of the assets of the banks declines below the value of liabilities which ultimately leads to insolvency.

Banking Failure (or Bank Runs): As the problems of the banks increase and banks start to go bankrupt, the depositors who do not want to lose their money, line up to withdraw their money from the banks. Though, some amount in deposits is insured by the governments, generally, it is very small. Banks do not have sufficient balances and they start to sell their assets. Asset prices are already low and it is hard to find buyers in the market. The selling of assets by the banks may lead to a rapid decline in asset prices. This is also known as **a fire sale**. Uncertainty about the health of the banks in the economy increases and people lose confidence in the safety of their deposits. This may lead to a full-fledged banking failure in the economy, i.e., several banks failing in a short duration.

Credit Shortage and Debt Deflation: When some banks fail, capital decreases as it becomes harder to get new capital as a result of low confidence in the market. Banks cut down on lending, and it becomes harder for the firms, especially the smaller and medium-sized firms, to get loans from the banks. Consumer credit also goes down as scrutiny by the banks increases. With the increase in the difficulties of the financial sector, the flow of credit declines in the economy. Decreases in the quantum of credit not only affect the new investment opportunities but also the operations of the firm which rely on debt for running their operation. A decrease in consumer credit leads to a decline in consumption demand. This leads to a decline in the economic activities in the country, a **recession**, and prices fall which increases the problems for the business firms. With the decrease in the prices, in real terms, the debt of the firms increases, known as **debt deflation**. Prices and incomes fall while the debt burden increases which leads to insolvency of the firms. This generates a kind of vicious circle. Deterioration in the health of the financial sector creates a shortage of credit that leads to an economic downturn which in turn decreases the profitability and stock prices of the firms. This may lead to insolvencies in the economy further deteriorating the health of the financial sector. In a financial crisis, several financial and non-financial firms collapse.

24.4.1 Case Study: The Great Recession (2007-2009)

From the great depression to the great recession, there were many recessionary periods in between. Out of these 2007 was the worst. Before this recession, there was the dot.com crisis in the US which was a crisis in the IT industry. To give a boost to this sector the US Federal Reserve decreased the rate of interest. Instead, this investment went to the real estate sector in the US. Soft loans were given for real estate properties. Multiple loans were advanced on the same mortgage. People who had poor credit history were also given loans. Banks who had these mortgages advanced financial instruments (shares, bonds, futures etc.) could not recover their dues. These financial instruments were even purchased by people sitting in other countries. There was a bubble in the real estate which was likely to burst. Ultimately, the prices in real estate started to fall. People defaulted on loans. Looking at the panic, people who had bought securities based on these mortgages started to exit. So, on one hand, people did not pay loans to banks and on the other hand the prices of their mortgage-based securities decreased. It was a double-edged sword for the banking system. The NPAs increased drastically. Major financial institutions collapsed in 2008. Lehman Brothers which was the 4th largest bank at that time, collapsed in September 2008. Ultimately the recession had spread to another sector too. Initially, India did not see any significant impact but eventually, the impact could be seen through the financial sector and decreased demand for Indian exports by the US and Europe. As a result of this recession, output decreased by 4.3%. Banks failed; housing prices declined by 31.8 per cent; unemployment peaked at 10% in October 2009 while it was 5% in December 2007; public debt increased, and protectionism increased.

24.5 FINANCIAL CRISES AND ECONOMIC CRISES

The financial system is an integral part of the economy and, both in good times and bad times, it amplifies economic activities. As financial development helps in economic growth a financial crisis worsens the economic problems. It is not necessary that a financial crisis would lead to an economic crisis; an economic crisis may also be a cause for the financial crisis. Finance is a key factor in capital formation and business activities.

When finance deteriorates it has a severe impact on economic activities. As we have discussed, during a financial crisis asset prices and price levels and at the same time, an increase in interest rates cause a decrease in the profitability of the firms. Credit crunch and a decline in profitability lead to a decrease in overall economic activities. The defaults and insolvency in non-financial firms would increase.

As a financial crisis looms a country's capital may fly to other countries in search of safer investment opportunities. It would have a twofold impact on the country. One, it would further deteriorate the credit crunch in the economy. Second, with a flight of capital exchange rate will be affected adversely which will impact the trade. Trade disturbance would aggravate the economic turbulences.

An economic crisis can also create financial crisis. An economic downturn would decrease the sales and profitability of the firms which would lead to an increase in defaults in the repayments of loans they have borrowed from financial institutions. This would hurt the balance sheets of the banks and other financial institutions may further lead to a financial crisis.